

Unrolled List
–Display List
–Program Analysis

Program

```
void displayList() {
    if (head == NULL) {
        cout << "The list is empty.\n";
        return;
    }

    struct Node* temp = head;
    int nodeCount = 1;

    cout << "\n--- Unrolled Linked List Content ---\n";
    while (temp != NULL) {
        cout << "Node " << nodeCount << " (" << temp->numElements <<
"/" << MAX_ELEMENTS << " elements): [";
        for (int i = 0; i < temp->numElements; i++) {
            cout << temp->elements[i];
            if (i < temp->numElements - 1) {
                cout << ", ";
            }
        }
        cout << "]\n";

        temp = temp->next;
        nodeCount++;
    }
    cout << "-----\n";
}

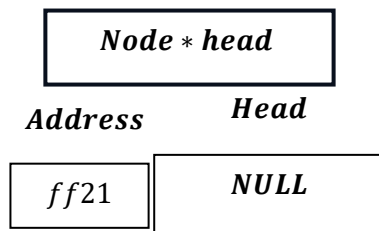
.....

    case 3:
        displayList();
        break;
```

Program Analysis

1. List is Empty

```
if (head == NULL) {  
    cout << "The list is empty.\n";  
    return;  
}
```

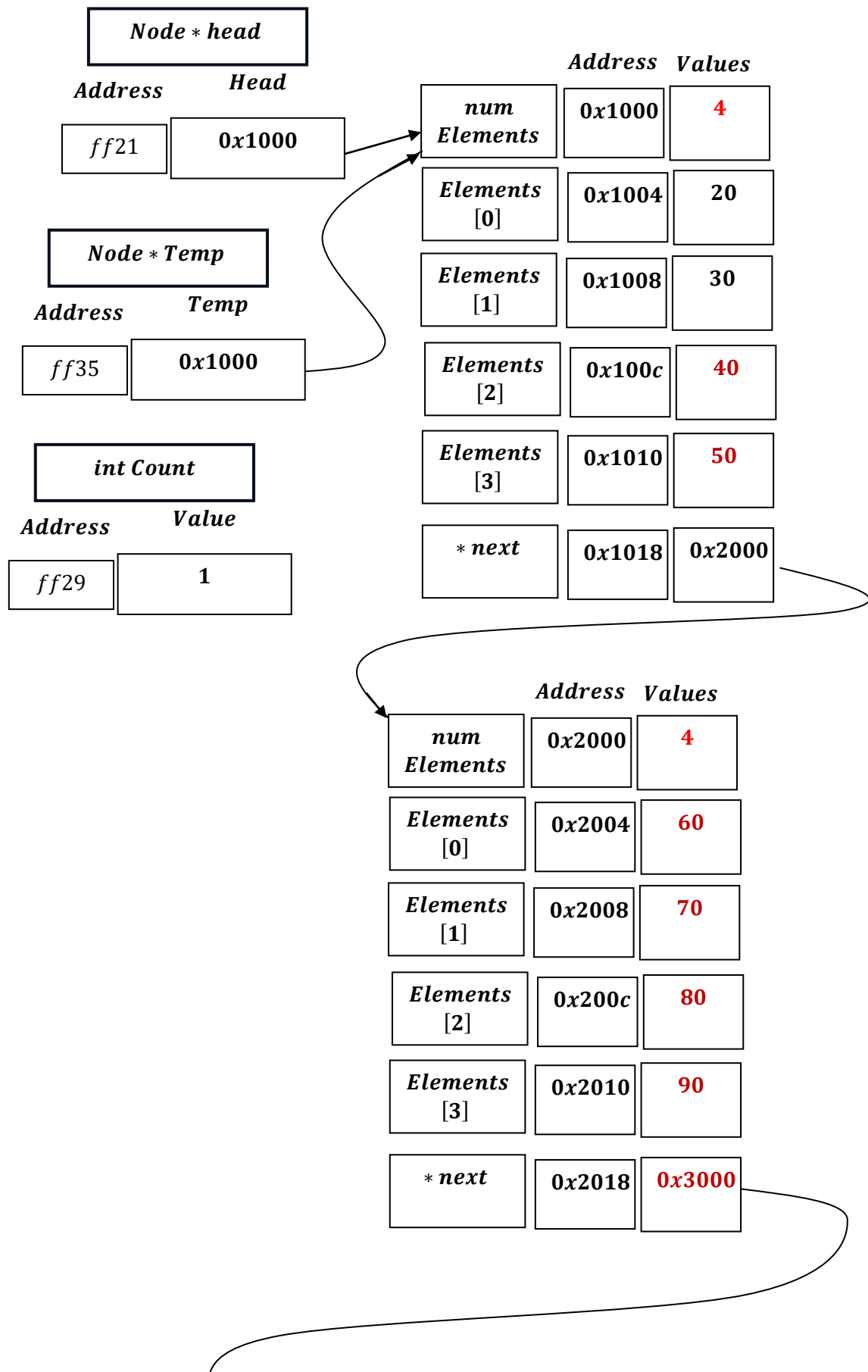


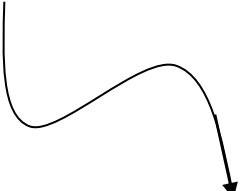
As head = NULL, it prints : `` The list is empty.``

Return void and exit from the function.

2. List is not Empty

```
struct Node* temp = head;  
int nodeCount = 1;
```





	<i>Address</i>	<i>Values</i>
<i>num Elements</i>	0x3000	4
<i>Elements [0]</i>	0x3004	100
<i>Elements [1]</i>	0x3008	110
<i>Elements [2]</i>	0x300C	120
<i>Elements [3]</i>	0x3010	130
<i>* next</i>	0x3018	NULL

```

cout << "\n--- Unrolled Linked List Content ---\n";
while (temp != NULL) {
    cout << "Node " << nodeCount << " (" << temp->numElements <<
"/" << MAX_ELEMENTS << " elements): [";
    for (int i = 0; i < temp->numElements; i++) {
        cout << temp->elements[i];
        if (i < temp->numElements - 1) {
            cout << ", ";
        }
    }
    cout << "]\n";

    temp = temp->next;
    nodeCount++;
}

```

Print `` --- Unrolled Linked List Content --- ``

While(Temp: 0x1000 $\neq$ NULL) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \end{array} \right]$

Print : Node 1 (temp $\rightarrow$ numElements: 4 \ MAX_ELEMENTS: 4 elements): [

for: int $i = 0$; $i < temp \rightarrow numElements: 4$

Print: temp $\rightarrow$ elements[0] := 20

if($i = 0 < temp \rightarrow numElements: 4 - 1 = 3$)

Print: `` , ``

$i = i + 1 = 0 + 1 = 1$

for: int $i = 1$; $i < temp \rightarrow numElements: 4$

Print: temp $\rightarrow$ elements[1] := 30

if($i = 1 < temp \rightarrow numElements: 4 - 1 = 3$)

Print: `` , ``

$i = i + 1 = 1 + 1 = 2$

for: int $i = 2$; $i < temp \rightarrow numElements: 4$

Print: temp $\rightarrow$ elements[2] := 40

if($i = 2 < temp \rightarrow numElements: 4 - 1 = 3$)

Print: `` , ``

$i = i + 1 = 2 + 1 = 3$

for: int $i = 3$; $i < temp \rightarrow numElements: 4$

Print: temp $\rightarrow$ elements[3] := 50

if($i = 3 < temp \rightarrow numElements: 4 - 1 = 3$) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

$i = i + 1 = 3 + 1 = 4$

for: int i = 4; i < temp → numElements: 4 $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

```
cout << "]\n";
```

Print: ``]`` and next : newline i.e. end of line and start from a newline.

And finally we have print:

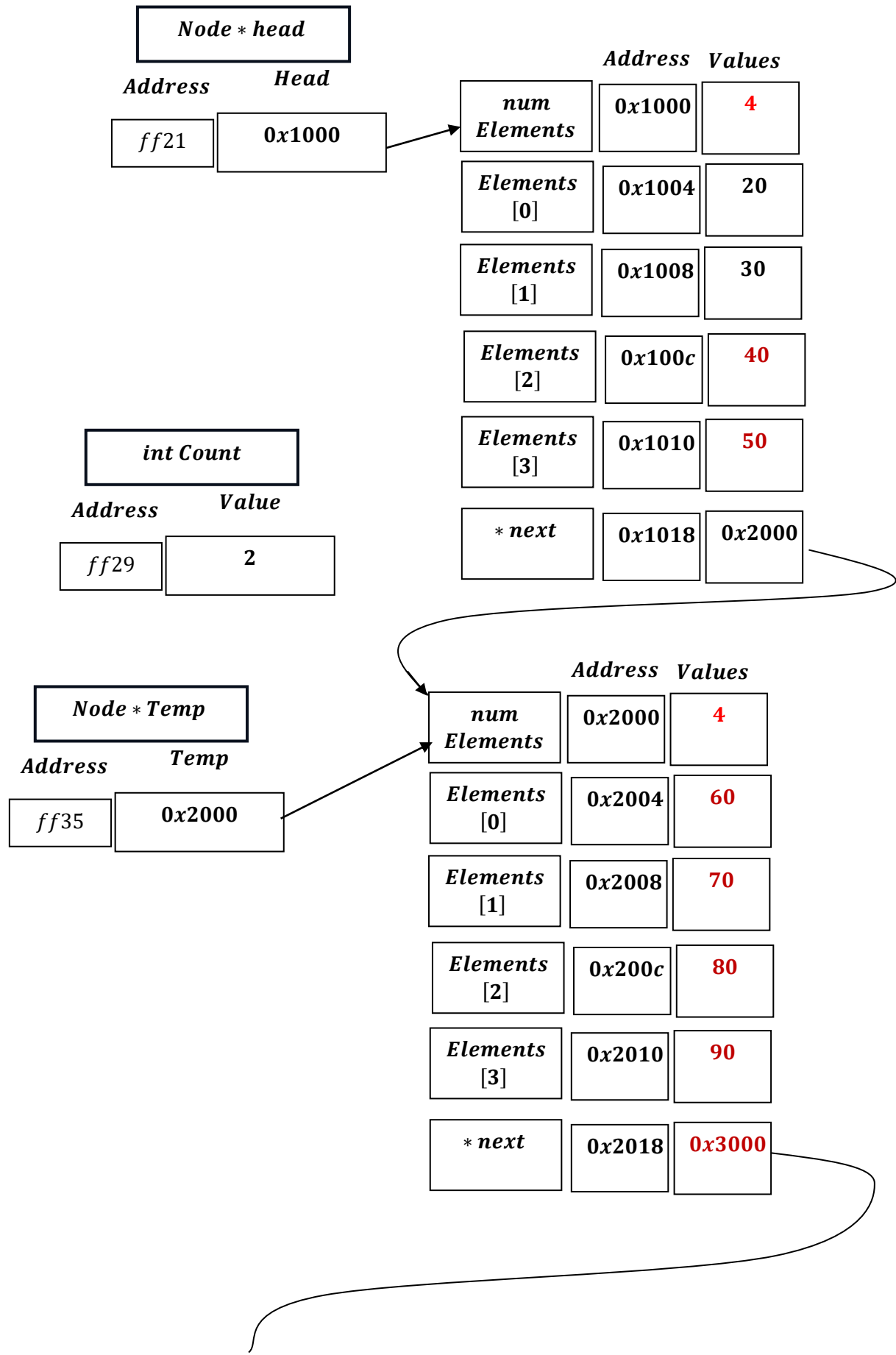
— — —Unrolled Linked List Content — — —

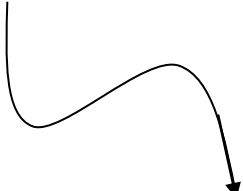
Node 1 (4 /4 elements): [20, 30, 40, 50]

```
temp = temp->next;  
nodeCount++;
```

Temp = Temp → Next : 0x2000

nodeCount = nodeCount + 1 = 1 + 1 = 2.





	<i>Address</i>	<i>Values</i>
<i>num Elements</i>	0x3000	4
<i>Elements [0]</i>	0x3004	100
<i>Elements [1]</i>	0x3008	110
<i>Elements [2]</i>	0x300C	120
<i>Elements [3]</i>	0x3010	130
<i>* next</i>	0x3018	NULL

While(*Temp*: 0x2000 $\neq$ *NULL*) $\left[\begin{array}{c} \textit{Condition} \\ \textit{is} \\ \textit{True} \end{array} \right]$

Print : *Node 2* (*temp* $\rightarrow$ *numElements*: 4 \ *MAX_ELEMENTS*: 4 *elements*): [

for: *int i* = 0; *i* < *temp* $\rightarrow$ *numElements*: 4

Print: *temp* $\rightarrow$ *elements*[0] := 60

if(*i* = 0 < *temp* $\rightarrow$ *numElements*: 4 – 1 = 3)

Print: `` , ``

i = *i* + 1 = 0 + 1 = 1

for: *int i* = 1; *i* < *temp* $\rightarrow$ *numElements*: 4

Print: *temp* $\rightarrow$ *elements*[1] := 70

if(*i* = 1 < *temp* $\rightarrow$ *numElements*: 4 – 1 = 3)

Print: `` , ``

i = *i* + 1 = 1 + 1 = 2

for: int i = 2; i < temp → numElements: 4

Print: temp → elements[2] := 80

if(i = 2 < temp → numElements: 4 - 1 = 3)

Print: `` , ``

i = i + 1 = 2 + 1 = 3

for: int i = 3; i < temp → numElements: 4

Print: temp → elements[3] := 90

if(i = 3 < temp → numElements: 4 - 1 = 3) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

i = i + 1 = 3 + 1 = 4

for: int i = 4; i < temp → numElements: 4 $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

```
cout << "]\n";
```

Print: `` ] `` and next : newline i.e. end of line and start from a newline.

And finally we have print:

— — Unrolled Linked List Content — —

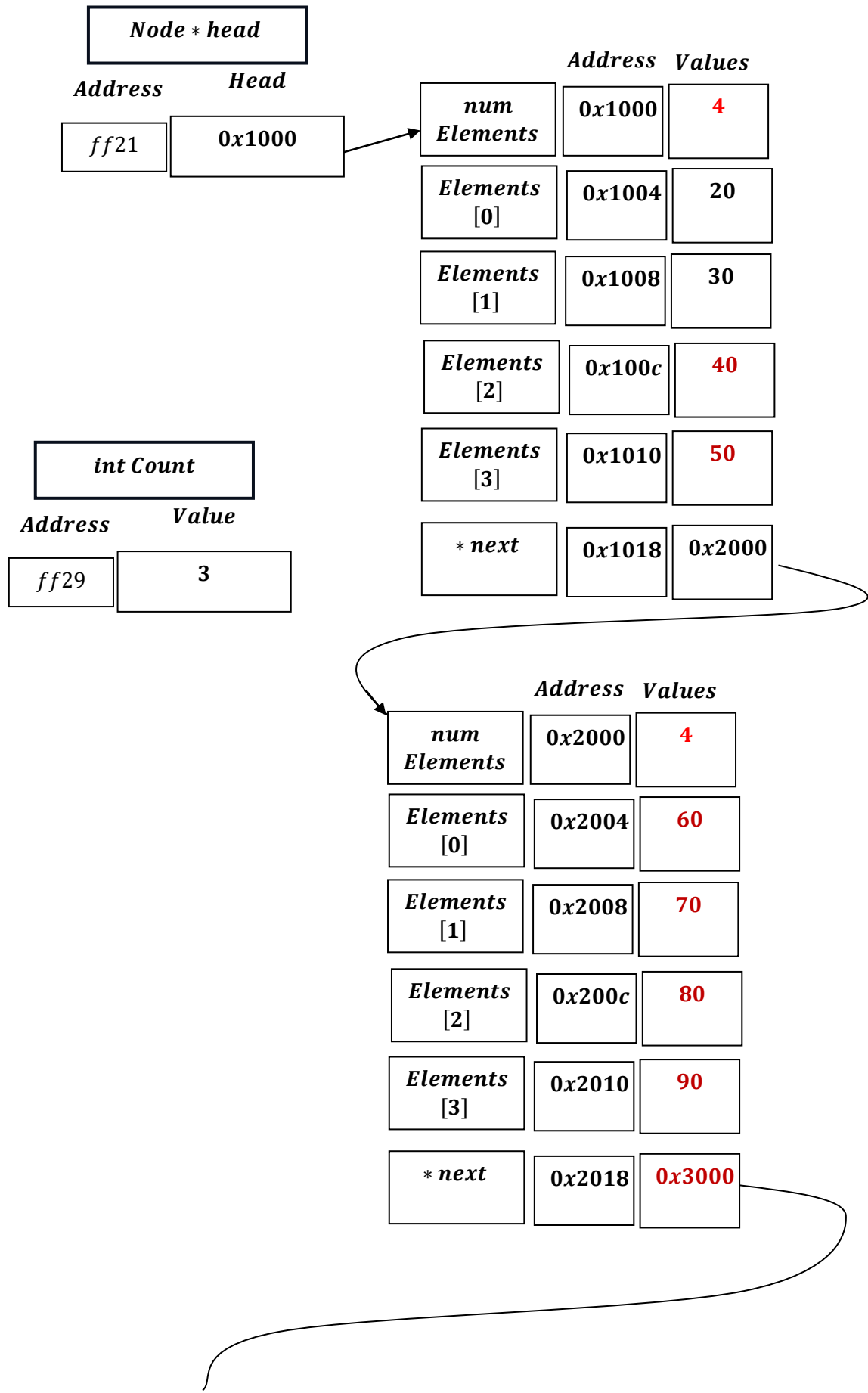
Node 1 (4 /4 elements): [20,30,40,50]

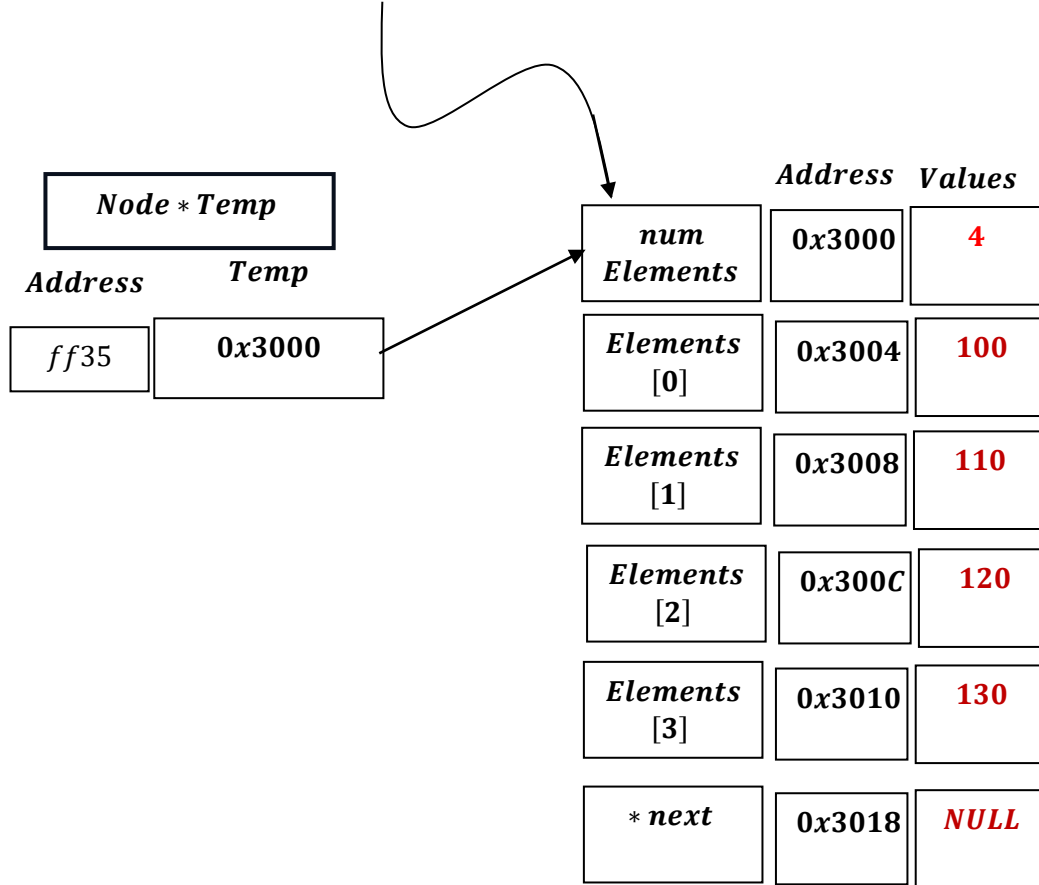
Node 2 (4 /4 elements): [60,70,80,90]

```
temp = temp->next;  
nodeCount++;
```

Temp = Temp → Next : 0x3000

nodeCount = nodeCount + 1 = 2 + 1 = 3.





While(Temp: 0x3000 $\neq$ NULL) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{True} \end{array} \right]$

Print : Node 3 (temp $\rightarrow$ numElements: 4 \ MAX_ELEMENTS: 4 elements): [

for: int $i = 0; i < temp \rightarrow numElements: 4$

Print: temp $\rightarrow$ elements[0] := 100

if($i = 0 < temp \rightarrow numElements: 4 - 1 = 3$)

Print: `` , ``

$i = i + 1 = 0 + 1 = 1$

for: int $i = 1; i < temp \rightarrow numElements: 4$

Print: temp $\rightarrow$ elements[1] := 110

if($i = 1 < temp \rightarrow numElements: 4 - 1 = 3$)

Print: `` , ``

$i = i + 1 = 1 + 1 = 2$

for: int i = 2; i < temp → numElements: 4

Print: temp → elements[2] := 120

if(i = 2 < temp → numElements: 4 – 1 = 3)

Print: `` , ``

i = i + 1 = 2 + 1 = 3

for: int i = 3; i < temp → numElements: 4

Print: temp → elements[3] := 130

if(i = 3 < temp → numElements: 4 – 1 = 3) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

i = i + 1 = 3 + 1 = 4

for: int i = 4; i < temp → numElements: 4 $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

```
cout << "]\n";
```

Print: `` ] `` and next : newline i.e. end of line and start from a newline.

And finally we have print:

– – –Unrolled Linked List Content – – –

Node 1 (4 /4 elements): [20, 30, 40, 50]

Node 2 (4 /4 elements): [60, 70, 80, 90]

Node 3 (4 /4 elements): [100, 110, 120, 130]

```
temp = temp->next;  
nodeCount++;
```

Temp = Temp → Next : NULL

nodeCount = nodeCount + 1 = 3 + 1 = 4.

While(Temp: NULL $\neq$ NULL) $\left[\begin{array}{c} \text{Condition} \\ \text{is} \\ \text{False} \end{array} \right]$

And , then we print:

```
cout << "-----\n";
```

Therefore , final output:

— — —Unrolled Linked List Content — — —

Node 1 (4 /4 elements): [20,30,40,50]

Node 2 (4 /4 elements): [60,70,80,90]

Node 3 (4 /4 elements): [100,110,120,130]
